

VIP Architecture: kube-vip and MetalLB

Two separate systems handle virtual IPs in this cluster, at different layers and for different purposes. They share the same L2/ARP mechanism but serve entirely different roles.

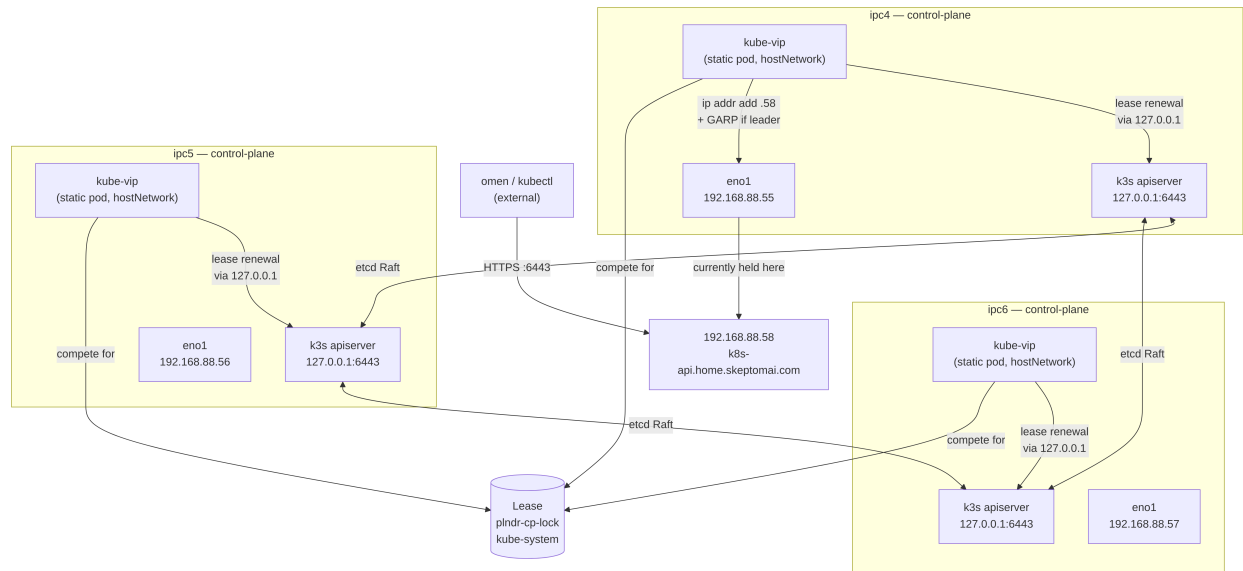
kube-vip — Control-Plane HA VIP

Why it's different from a normal controller

Most Kubernetes controllers are Deployments or DaemonSets — they are *scheduled* by the control plane and depend on the API server being reachable. kube-vip cannot work that way because it *is* the thing that makes the API server reachable from outside. It solves a bootstrapping problem.

kube-vip runs as a **static pod**: a manifest written to `/var/lib/rancher/k3s/agent/pod-manifests/` on each control-plane node. The **kubelet** reads that directory directly at startup and launches the pod without consulting the API server. kube-vip is therefore running *before* etcd is healthy, *before* the API server accepts connections, and *before* any scheduler exists.

Architecture



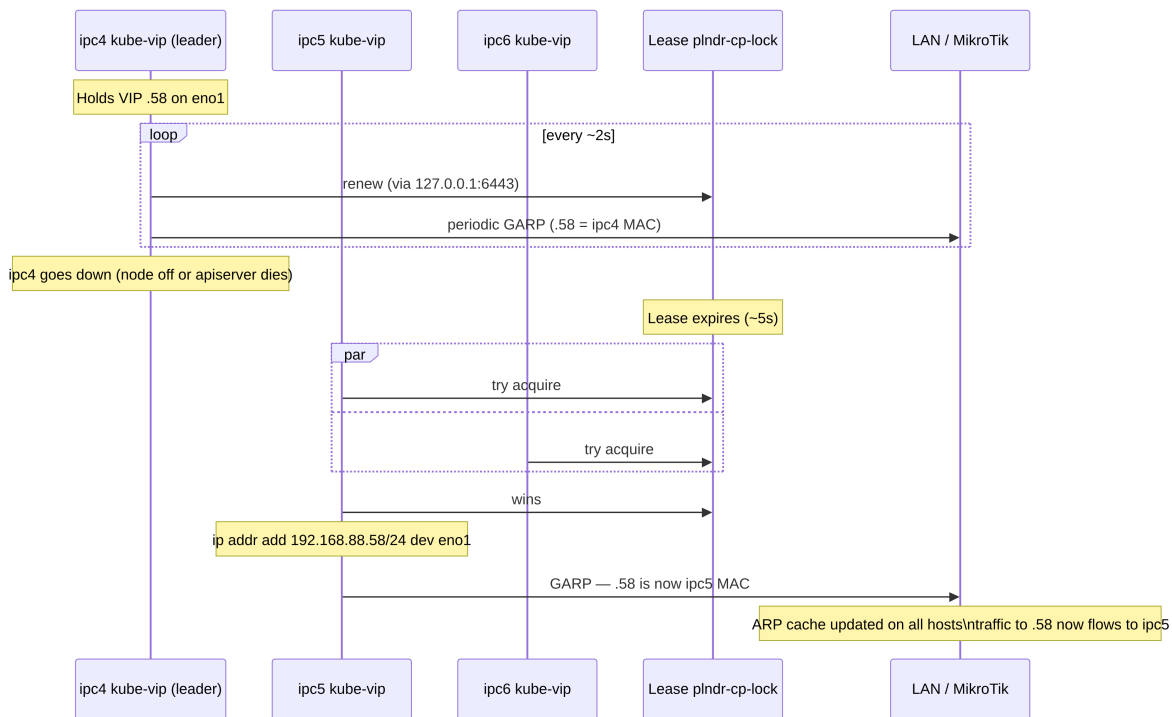
The local-apiserver trick

kube-vip's leader election uses a Kubernetes Lease object — but which API server does it talk to? If it used the VIP itself (.58:6443), a dead local apiserver wouldn't affect the lease, and the VIP would stay stuck on a broken node. Instead, kube-vip is configured with:

```
env:
- name: KUBERNETES_SERVICE_HOST
  value: "127.0.0.1"
- name: KUBERNETES_SERVICE_PORT
  value: "6443"
```

Each kube-vip pod talks *only* to its own node's apiserver on 127.0.0.1. If that apiserver dies, the local kube-vip can't renew the lease → it relinquishes leadership → a node with a live apiserver takes over and the VIP floats.

Failover sequence



The MAC address of **.58** **changes** on failover — this is the key difference from VRRP, which keeps a constant virtual MAC. The gratuitous ARP is essential: without it every device on the LAN would keep sending traffic to ipc4's (now silent) MAC until their ARP cache timed out naturally.

What kube-vip does NOT do

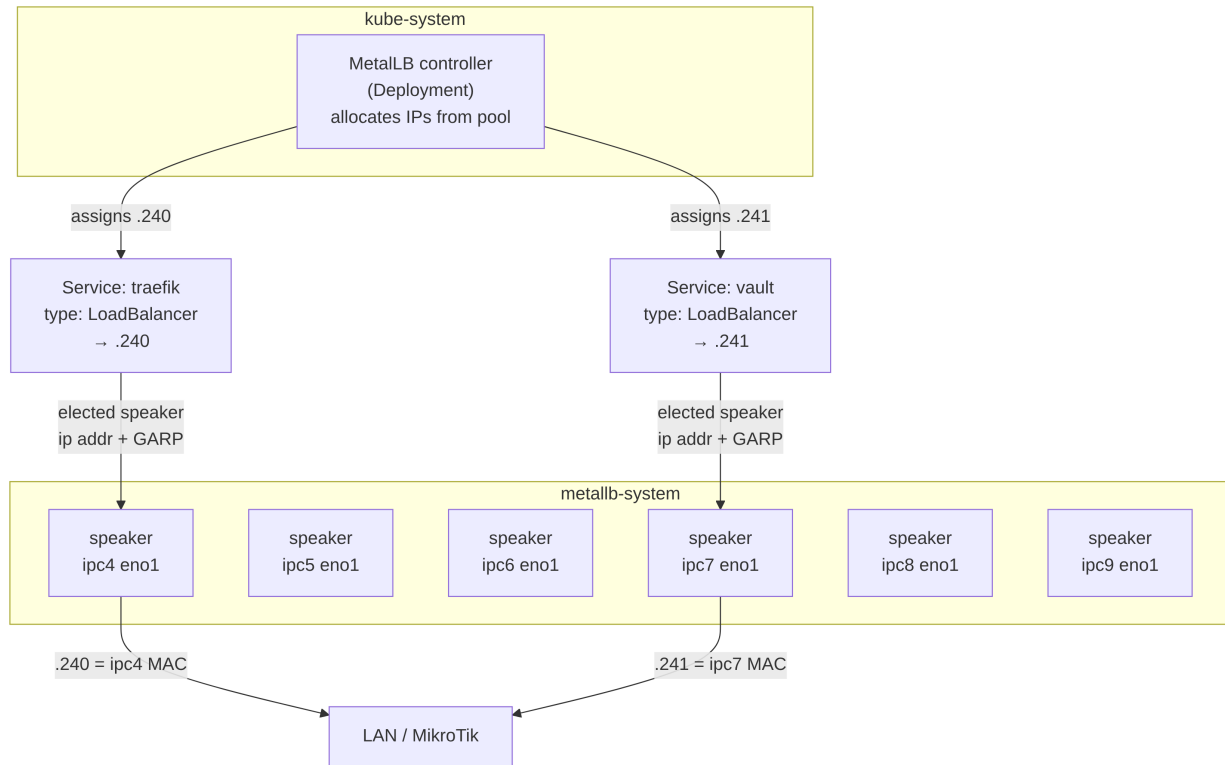
kube-vip is configured with `svc_enable=false` — it does not handle `type: LoadBalancer` Services. That is MetalLB's job. kube-vip only manages the single control-plane VIP.

MetalLB — Workload LoadBalancer VIPs

Architecture

MetalLB is a standard Kubernetes controller — it runs as a normal Deployment and DaemonSet, scheduled and managed by Kubernetes, with no bootstrapping concerns.

- **controller** (Deployment): watches for `type: LoadBalancer` Services and allocates IPs from the pool 192.168.88.240–.250.
- **speaker** (DaemonSet, `hostNetwork: true`): runs on all 6 nodes. For each assigned IP, an elected speaker plumbs it onto its own `eno1` and answers ARP — same mechanism as kube-vip, different scope.



Each Service gets its own elected speaker — different Services can be announced from different nodes simultaneously.

Pool exhaustion isn't the workload limit

The pool .240-.250 (11 IPs) does **not** cap the number of workloads you can run. It caps the number of distinct external IPs. Traefik consumes one (.240) and acts as a multiplexer for all HTTP/HTTPS traffic — dozens of web services share that single IP. The remaining pool IPs are available for services that genuinely need a dedicated external IP (raw TCP, UDP, non-HTTP protocols that can't share an ingress).

Traefik — L7 Routing Behind .240

Traefik is the ingress controller. It holds a single `type: LoadBalancer` Service at .240, receives all incoming HTTP/HTTPS traffic, and routes each request to the correct backend based on the HTTP `Host` header (and optionally path).

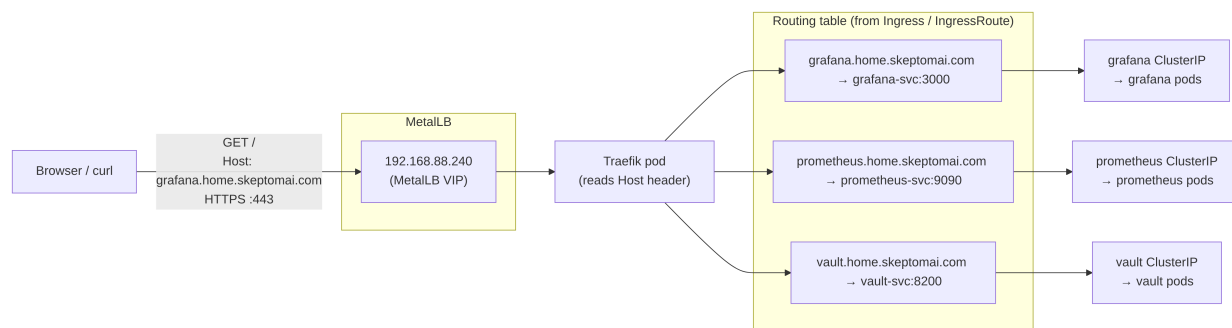
How Traefik knows where to route

Traefik watches the Kubernetes API for two kinds of routing rules:

1. **Ingress resources** (standard Kubernetes): specify host + path → Service name + port. Kubernetes-native, works with any ingress controller.
2. **IngressRoute CRDs** (Traefik-specific): richer routing — regex paths, middleware chains, TLS options, TCP routes — expressed as Traefik's own custom resources.

When you create either, Traefik builds an internal routing table and starts forwarding matching traffic immediately — no restart required.

Traffic flow



The flow for a single request: 1. Client resolves `grafana.home.skeptomai.com` → `192.168.88.240` (DNS) 2. TCP connection arrives at `.240:443`; MetalLB delivers it to the node whose speaker holds `.240` (currently `ipc4`) 3. Traefik terminates TLS, reads the decrypted `Host:` header 4. Matches against routing table → `grafana-svc:3000` 5. Proxies to the Grafana ClusterIP Service, which load-balances across Grafana pods

Traefik knows nothing about which node the pods are on — that's the ClusterIP Service's job (backed by kube-proxy iptables rules on every node).

TLS Termination and Certificate Management

The problem Traefik has to solve

Traefik can terminate TLS for any hostname it receives traffic for, but it needs a certificate valid for that name. The routing problem (which backend?) and the certificate problem (where does the cert come from?) are completely separate.

Where certs come from — options

Built-in ACME (Let's Encrypt) Traefik has a built-in ACME client that can automatically obtain and renew certs. Two challenge types: - HTTP-01: Let's Encrypt verifies domain ownership by hitting `/.well-known/acme-challenge/` on port 80. Requires public inbound access — doesn't work for internal names like `grafana.home.skeptomai.com`. - DNS-01: Traefik writes a TXT record via your DNS provider's API to prove ownership. Works for internal names and wildcard certs with no public exposure required. Rate-limited by Let's Encrypt.

Traefik stores ACME certs in a local JSON file (`acme.json`). In Kubernetes this is problematic because pods are ephemeral — you need a PersistentVolume or you'll burn through Let's Encrypt rate limits on every restart.

cert-manager (the standard Kubernetes approach) cert-manager is a separate controller that manages the full certificate lifecycle and stores results as Kubernetes TLS Secrets. Traefik reads those Secrets; it doesn't know or care how cert-manager created them. cert-manager supports multiple issuers:

Issuer type	How it works	Good for
SelfSigned CA	Generates a self-signed cert Signs with a cert+key in a Kubernetes Secret	bootstrap only internal CA
ACME	Gets certs from Let's Encrypt (HTTP-01 or DNS-01)	public hostnames
Vault	Signs via Vault PKI secrets engine	private CA with audit trail

Default fallback If no cert matches the incoming hostname, Traefik falls back to a built-in self-signed default cert — causing browser TLS warnings. This is the bare Traefik install state.

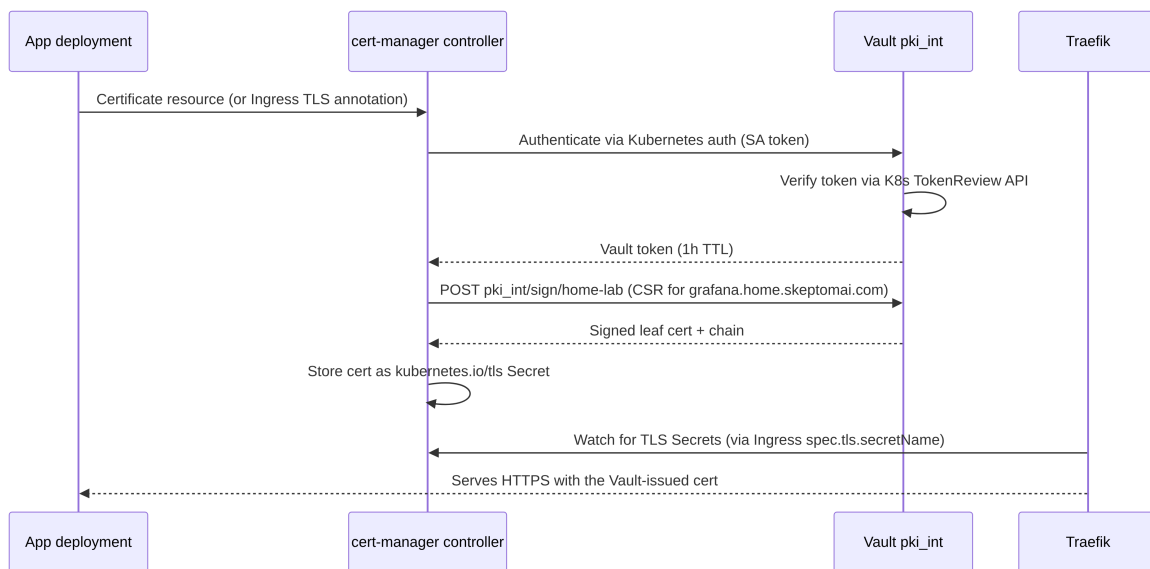
Certificate chain in this cluster

This cluster uses a **Vault PKI intermediate CA** under a **self-signed internal root**:

```
internal-ca (self-signed, cert-manager manages, 10y lifetime)
├─ Vault pki_int (intermediate CA, signed by internal-ca, 5y lifetime)
│   └─ Leaf certs (issued by Vault PKI role 'home-lab', 1y max)
│       - *.home.skeptomai.com
│       - *.svc.cluster.local
```

Why this hierarchy: - The internal-ca root is a single cert you install once in your browser/OS trust store. Every cert in the cluster is then automatically trusted — no per-service warnings. - Vault PKI manages the intermediate CA key inside Vault's encrypted storage, with full audit logging of every cert issuance. cert-manager asks Vault to sign certs; the private key for the intermediate CA never leaves Vault. - The intermediate CA layer means you can rotate the Vault PKI without touching the root cert that's installed in browsers.

How a certificate gets issued (request flow)



Vault Kubernetes auth — how it works

cert-manager authenticates to Vault without a static token:

1. cert-manager requests a short-lived token for its own ServiceAccount via the Kubernetes TokenRequest API.
2. It sends that token to Vault's `auth/kubernetes/login` endpoint.
3. Vault calls the Kubernetes TokenReview API to verify the token is valid and belongs to the `cert-manager` SA in the `cert-manager` namespace.
4. If valid, Vault returns a short-lived Vault token (1h) bound to the `cert-manager-pki` policy.
5. cert-manager uses that token to call `pki_int/sign/home-lab` and get the cert signed.

No static Vault tokens are stored anywhere in Kubernetes.

Requesting a cert for an Ingress

```
# Option A: Ingress annotation (cert-manager creates the Certificate automatically)
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: grafana
  annotations:
    cert-manager.io/cluster-issuer: vault-pki-issuer
spec:
  tls:
    - hosts: [grafana.home.skeptomai.com]
      secretName: grafana-tls # cert-manager fills this
  rules:
    - host: grafana.home.skeptomai.com
    ...
```

```
# Option B: Explicit Certificate resource (more control over renewal timing, etc.)
apiVersion: cert-manager.io/v1
kind: Certificate
metadata:
  name: grafana-tls
  namespace: monitoring
spec:
  secretName: grafana-tls
  duration: 8760h
  issuerRef:
    name: vault-pki-issuer
    kind: ClusterIssuer
  dnsNames:
    - grafana.home.skeptomai.com
```

Setup

Vault PKI must be configured before the ClusterIssuer will become Ready:

```
bash scripts/setup-vault-pki.sh
kubectl apply -k manifests/vault-pki
kubectl get clusterissuer vault-pki-issuer
```

The setup script: 1. Enables `pki_int` in Vault 2. Generates the intermediate CA CSR (private key stays in Vault) 3. Signs the CSR via cert-manager's `internal-ca-issuer` 4. Imports the signed cert chain back into Vault 5. Creates the `home-lab` role and Kubernetes auth configuration

Does SPIRE play any role here?

No — and it's worth being precise about why.

SPIRE issues **X.509 SVIDs**: short-lived certificates with SPIFFE URI SANs (`spiffe://ipc.local/ns/foo/sa/bar`), not DNS SANs. SVIDs are workload *identity* credentials used for pod-to-pod mTLS — they are not browser-trusted and are not suitable for HTTP ingress TLS.

In this cluster, SPIRE uses its own self-managed CA (trust domain `ipc.local`), entirely separate from the Vault PKI / `internal-ca` chain. The two trust hierarchies do not intersect:

	SPIRE SVIDs	Ingress TLS (Vault PKI)
Purpose	Pod-to-pod mTLS (workload identity)	Browser HTTPS (ingress TLS)
SANs	SPIFFE URI (<code>spiffe://ipc.local/...</code>)	DNS names (<code>*.home.skeptomai.com</code>)
Lifetime	Minutes to hours (auto-rotated by SPIRE)	Days to a year (cert-manager managed)
Trust root	SPIRE's own CA (<code>ipc.local</code>)	internal-ca (self-signed)
Browser trusted?	No	Yes, if internal-ca is installed

The integration point that *could* exist (but isn't configured here): Vault can be SPIRE's upstream authority, making Vault PKI the root for both SVID issuance and ingress TLS. In that scenario a single Vault PKI root would cover everything and you'd only need one cert in your browser trust store. This is a valid future evolution but adds operational complexity.

Side-by-side comparison

	kube-vip	MetalLB
Purpose	Control-plane HA	Workload LoadBalancer Services
VIP(s)	One: <code>192.168.88.58</code>	Many: <code>192.168.88.240-.250</code>
Runs as	Static pod (kubelet, no scheduler)	Deployment + DaemonSet (normal)
hostNetwork	Yes	Yes (speaker only)
ARP mechanism	GARP from real MAC	GARP from real MAC
MAC on failover	Changes	Changes
vs. VRRP	No virtual MAC	No virtual MAC
Leader election	Kubernetes Lease (via <code>127.0.0.1</code>)	Per-Service elected speaker
Cluster dependency	None — predates the cluster	Full — requires running cluster
L7 routing	No	No (MetalLB is L3/L4 only — Traefik does L7)